(ii) $\int e^{-5x} \sin 3x \, dx$

Using the formula

$$\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2+b^2}[a \sin bx - b \cos bx] + c$$

for $a = -5, b = 3$, we get

$$\int e^{-5x} \sin 3x \, dx = \left(\frac{e^{-5x}}{(-5)^2+3^2}\right)(-5 \sin 3x - 3 \cos 3x) + c$$

$$\int e^{-5x} \sin 3x \, dx = -\left(\frac{e^{-5x}}{34}\right)(5 \sin 3x + 3 \cos 3x) + c$$

EXERCISE 11.8

Integrate the following with respect to x

(1) (i) $e^{ax} \cos bx$ (ii) $e^{2x} \sin x$ (iii) $e^{-x} \cos 2x$

(2) (i) $e^{-3x} \sin 2x$ (ii) $e^{-4x} \sin 2x$ (iii) $e^{-3x} \cos x$

Result 11.2

$$\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$$

Proof

$$\text{Let } I = \int e^x [f(x) + f'(x)] dx$$

$$= \int e^x f(x) dx + \int e^x f'(x) dx$$

Take $u = f(x)$; $du = f'(x)\,dx$, in the first integral

$dv = e^x$; $v = e^x$,

That is, $I = e^x f(x) - \int e^x f'(x) dx + \int e^x f'(x) dx + c$

Therefore, $I = e^x f(x) + c$.

Examples 11.37

Evaluate the following integrals

(i) $\int e^x \left(\frac{1}{x} - \frac{1}{x^2}\right) dx$ (ii) $\int e^x (\sin x + \cos x)\, dx$ (iii) $\int e^x \left(\frac{1-x}{1+x^2}\right)^2 dx$

Solution

(i) $$\text{Let } I = \int e^x \left(\frac{1}{x} - \frac{1}{x^2}\right) dx$$

$$\text{Take } f(x) = \frac{1}{x}, \text{ then } f'(x) = \frac{-1}{x^2}$$

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